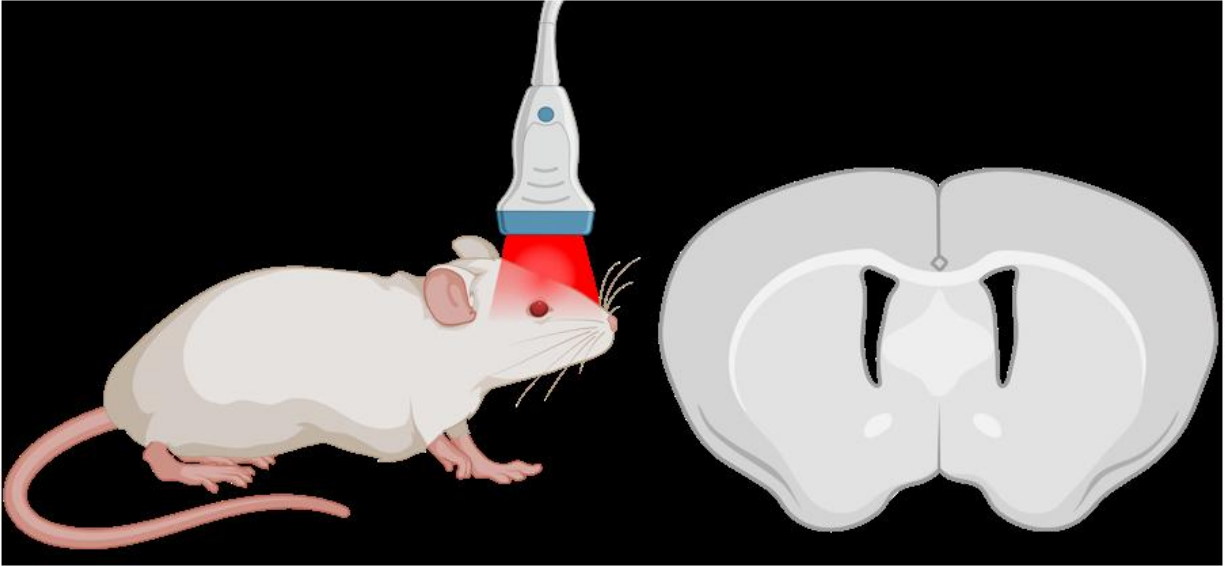


HUMoR fUSI Studio

User Manual

Data Analysis Pipeline



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Prepared from the uploaded HUMoR Analysis Tool ZIP and existing SOP

Generated: 30 April 2026

Scope note: This manual is focused on the data analysis workflow. Experimental preparation and scanning are summarized only where they affect data interpretation. For surgery, animal welfare and acquisition details, use the approved SOP and local protocol documentation.

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1. What this manual teaches

This user manual explains how to analyze functional ultrasound imaging data with HUMoR (Hemodynamic Ultrasound Imaging of Molecular Reporters) fUSI Studio. It is written for a new user who has a raw fUSI dataset and wants to move from data loading to QC, preprocessing, PSC/SCM visualization, ROI export, atlas registration, group analysis, and optional functional connectivity.

By the end of the manual, the user should be able to launch the GUI, load a 2D, step-motor or matrix/3D dataset, verify TR and timing, inspect data quality, apply only justified preprocessing, generate signal-change maps, validate masks/underlays, export ROI/time-course outputs, and prepare group-level summaries.

Tip / Trick: Do not treat the GUI as a black box. Every preprocessing step changes the dataset. Keep the raw dataset untouched, use the dataset dropdown carefully, and export the Studio Log at the end of each analysis session.

Use Windows! Unix is not supported!

2. HUMoR Studio

The main entry point is fusi_studio.m. The GUI is organized into seven user-facing sections: Dataset, Quality Control & Data Overview, Recommended Processing Steps, Advanced Processing, Visualization, Coregistration, and Advanced Analysis.

GUI section	Buttons	Purpose
1. Dataset	Load fUSI Data and dataset dropdown	Start here. Loads raw or processed datasets and controls the active dataset.
2. Quality Control & Data Overview	Full QC, Specific QC	Diagnose data stability, frame artifacts, temporal/spatial quality and reliability.
3. Recommended Processing Steps	Frame Rejection, Imregdemons, Scrubbing, Motor	Use when QC shows an issue or when the acquisition mode requires reconstruction or if needed for the type of data you have.
4. Advanced Processing	Temporal Smoothing/Subsampling, Filtering, PCA / ICA, Despiking	Use only when scientifically justified. Document settings.
5. Visualization	Time-Course Viewer, SCM, Video & SCM Mask, Mask Editor	Inspect raw/PSC dynamics, create maps, draw/load masks and export figures.
6. Coregistration	Registration to Atlas, Segmentation	Register anatomy/underlay to Allen atlas.
7. Advanced Analysis	Functional Connectivity, Group Analysis	Use after individual datasets pass QC and exported inputs are checked.

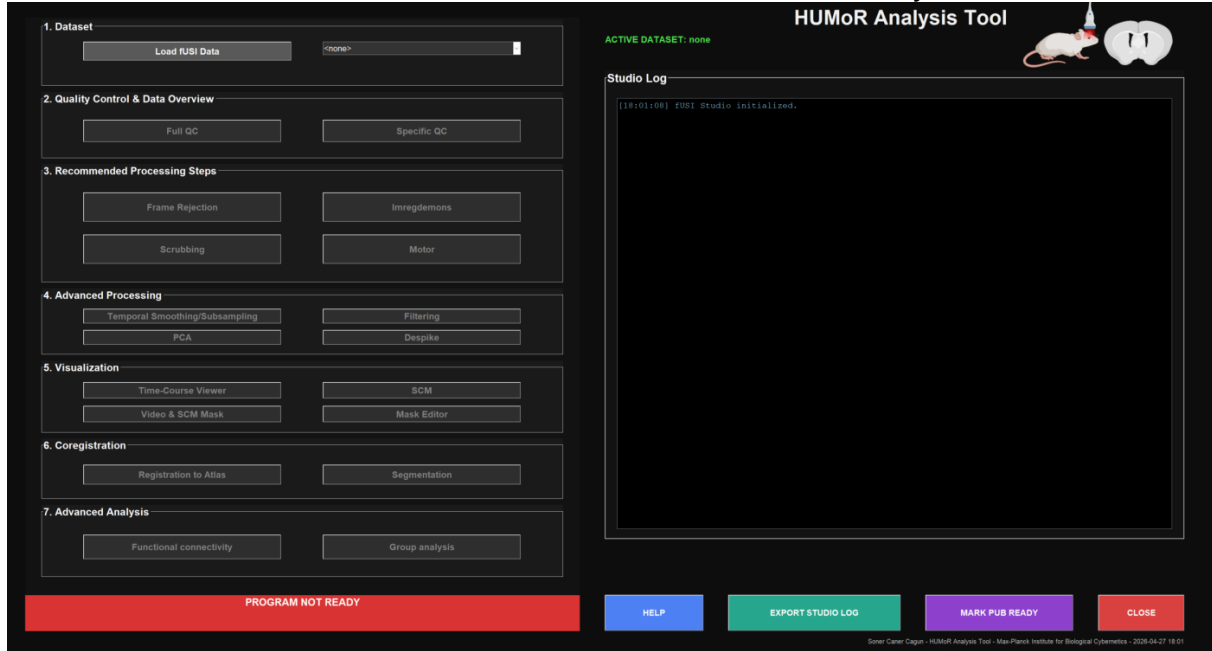


Figure 1: Main HUMoR Analysis Tool layout from the SOP screenshot.

3. Concepts used in the pipeline

Concept	Meaning in HUMoR Studio
fUSI signal	Power Doppler-like intensity changes from moving blood signal. Interpret as hemodynamic/CBV-like, not direct neuronal activity.
TR	Repetition time between fUSI frames/volumes. The loader and GUI ask the user to confirm it. Common defaults in the current workflow are 0.320 s for standard 2D and 0.480 s for matrix/3D probe data. This can be changed. You can also manually set the TR through a pop-up.
Baseline window	Time interval used to compute baseline intensity for PSC. Must be stable and before the stimulus/injection or major movement. You can also change the values dynamically.
PSC	Percentage signal change: $100 \times (\text{signal} - \text{baseline}) / \text{baseline}$. Used for SCM overlays and ROI time courses.
Underlay	Anatomical background/reference image below the PSC overlay; may be mean, median, external file, atlas underlay, or standardized Doppler-equalized image.
Overlay / signal mask	Mask that restricts where PSC/signal is displayed.
Brain / underlay mask	Mask that restricts anatomical underlay or valid brain area.
Active dataset	The dataset currently selected in the dropdown. Most mistakes happen when this is not the intended version.

3.1 Core calculations

For a voxel or pixel with intensity $I(t)$, the baseline B is computed from the selected baseline frames. PSC is then computed as:

$$\text{PSC}(t) = ((I(t) - B) / B) \times 100$$

The same logic applies to 2D time series $[Y \times T]$ and multi-slice/matrix data $[Y \times Z \times T]$. ROI time courses are generated by averaging PSC values inside the selected ROI/mask at each time point.

Tip / Trick: A bad baseline contaminates every PSC frame. Before exporting maps, inspect the raw movie, global signal, and ROI time course around the baseline window.

4. Installation and setup

1. Install MATLAB. The current code comments repeatedly target MATLAB 2017b and 2023b compatibility.
2. Install/enable required MATLAB toolboxes where available, especially Image Processing Toolbox for some registration and mask operations. Several helper functions include fallback behavior, but not every workflow can be fully toolbox-free.
3. Unzip or clone HUMOR-Analysis-Tool into a local working folder from here:
<https://github.com/SonerCgu/HUMOR-Analysis-Tool/tree/main>
4. Start MATLAB and add the HUMoR Analysis Tool folder and subfolders to the path. Keep old backup folders out of the active path when possible.
5. Confirm that allen_brain_atlas.mat is available if atlas registration or atlas-based segmentation is needed.
6. Run fusi_studio from the MATLAB Command Window.

5. Expected data and folder organization

The loader accepts MATLAB MAT files and NIfTI files. The preferred image variable is I, but loadFUSIData.m searches for variables such as I, IQR, img, image, data, volume, vol, scan, brain, func and other nested candidates. It keeps the last dimension as time.

Input type	Expected structure	Notes
.mat	[Y X T] or [Y X Z T]	Best supported route. Metadata fields such as md, metadata, TR, dt, Fs, time, par, baseline and mask can be passed through when present.
.nii	2D/3D/4D NIfTI converted by niftiread	Axes are permuted for MATLAB display. Verify dimensions and timing manually.
.nii.gz	Gunzip to temporary .nii then niftiread	Same caution as .nii; see developer note about TRWasImputed initialization.
Motor split files	Many slice/time MAT files	Use Motor reconstruction mode. Naming should be consistent, e.g. slice1_t001.mat.

5.1 Recommended folder structure

```

RawData/
  <project>/<animal>/<session>/<scan>.mat
AnalysedData/
  <project>/<animal>/<session>/<scan>/
    QC/
    Preprocessing/
    PSC/
    Visualization/
    Registration/
    Registration2D/
    FunctionalConnectivity/
    GroupAnalysis/
    Reports/

```

Tip / Trick: Never overwrite raw data. Every preprocessing step should create a new dataset version and the dropdown should be checked before running SCM, Video, Registration, or Group Analysis.

6. Step-by-step individual analysis workflow

Step 0: Prepare the analysis session

Open MATLAB in the HUMoR code folder. Add code folders to the path. Keep raw data read-only. Open your experiment log so TR, stimulus/injection timing and acquisition mode are available.

Step 1: Launch HUMoR Studio

Run `fusi_studio`. The main GUI appears with disabled buttons until data are loaded. The Studio Log is the main text record of actions.

Step 2: Load fUSI Data

Click Load fUSI Data and select the raw .mat, .nii or .nii.gz file. Confirm probe type and TR in the prompt. Check the active dataset label at the top.

Step 3: Run QC before preprocessing - Optional

Run Full QC for normal datasets. For large data, use Specific QC modules. Review saved QC images before modifying data.

Step 4: Apply only justified recommended processing

Use Frame Rejection or Scrubbing for bad frames, Imregdemons for drift/spatial instability, and Motor only for step-motor/multi-slice reconstruction. This highly depends on the type of data you are having.

Step 5: Use advanced processing only when needed

Temporal smoothing, filtering, PCA/ICA and despiking can improve readability but can also remove true biological signal. Save and report the settings. Make sure to always remove same components amongst different animals.

Step 6: Inspect with Time-Course Viewer

Use the live viewer to inspect raw or PSC dynamics, signal stability and obvious motion before building maps. Moreover, to can run the raw movie and see if there is motion.

Step 7: Create SCM maps

Open SCM. Select baseline, underlay, signal polarity, caxis, alpha modulation, smoothing, slice and masks. Export ROI/time-course and map outputs.

Step 8: Verify dynamic behavior with Video & SCM Mask

Open the video GUI to confirm that SCM findings are stable over time and not driven by a single bad frame.

Step 9: Draw/load masks if needed

Use Mask Editor to draw underlay/brain masks and overlay/signal masks. Save mask bundles with clear names. The name assignment is done automatically.

Step 10: Register to atlas when individual data are clean

Use Registration to Atlas after a good anatomical underlay exists. Visually inspect registration before trusting atlas labels.

Step 11: Export and log

Export Studio Log, ROI text files, figures, SCM PPT if used, and mark publication-ready only after manual review of data.

Based on the version you are using the GUIs might look different, new functions might have been added or removed.

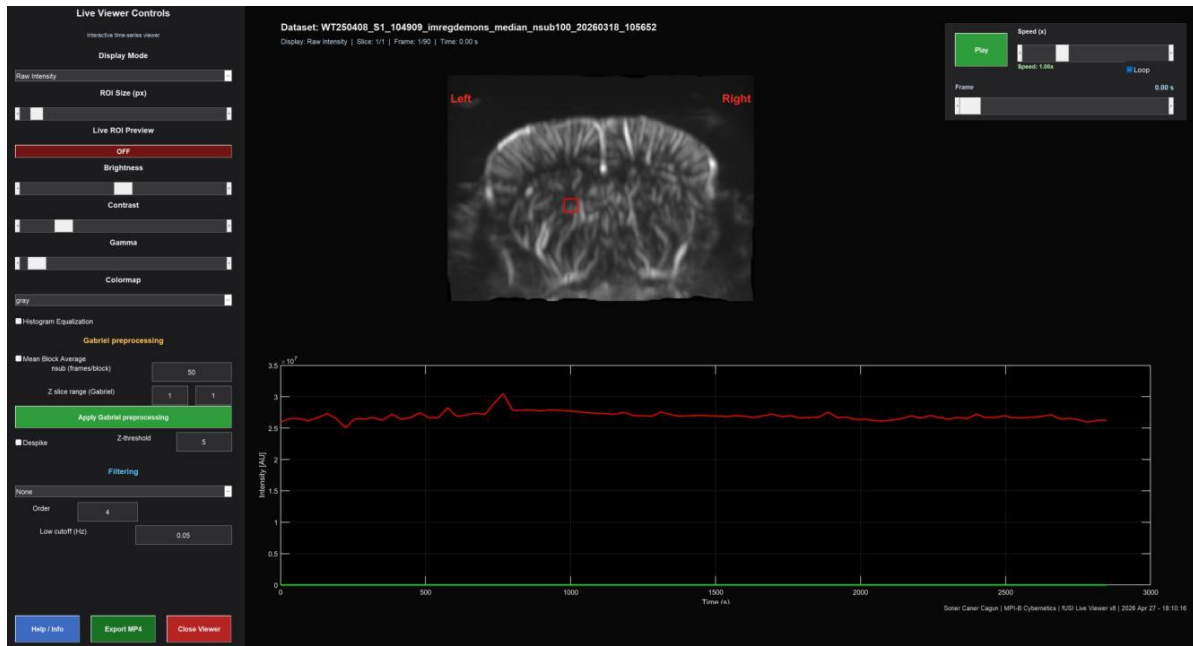


Figure 2: Time-course viewer example from SOP screenshot

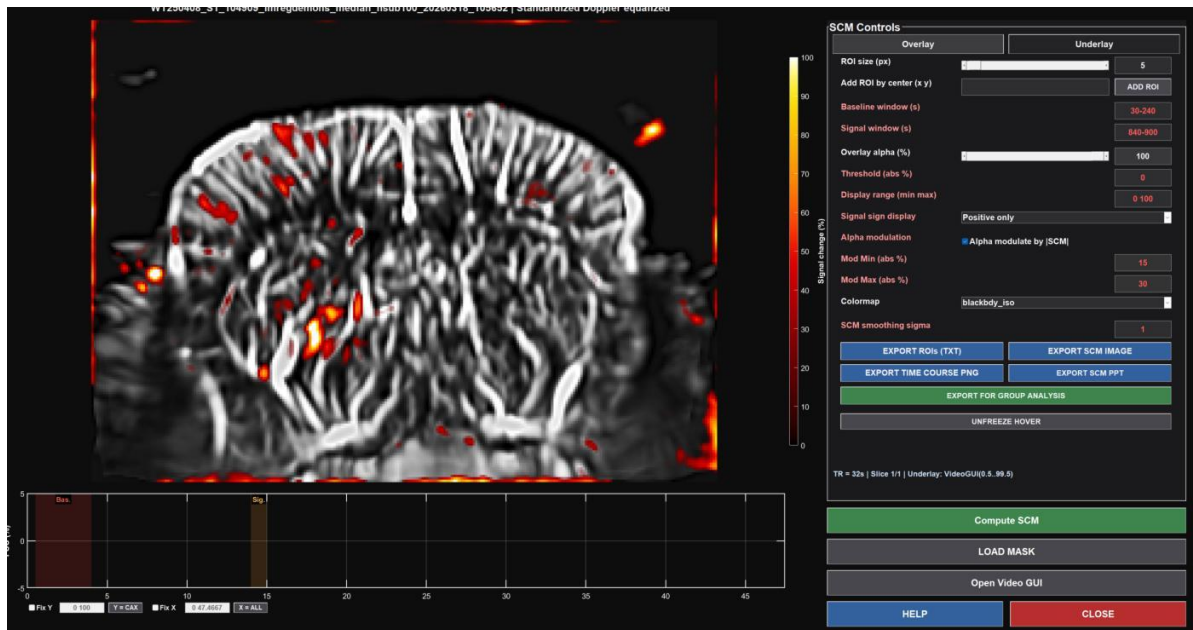


Figure 3: SCM GUI example from SOP screenshot.

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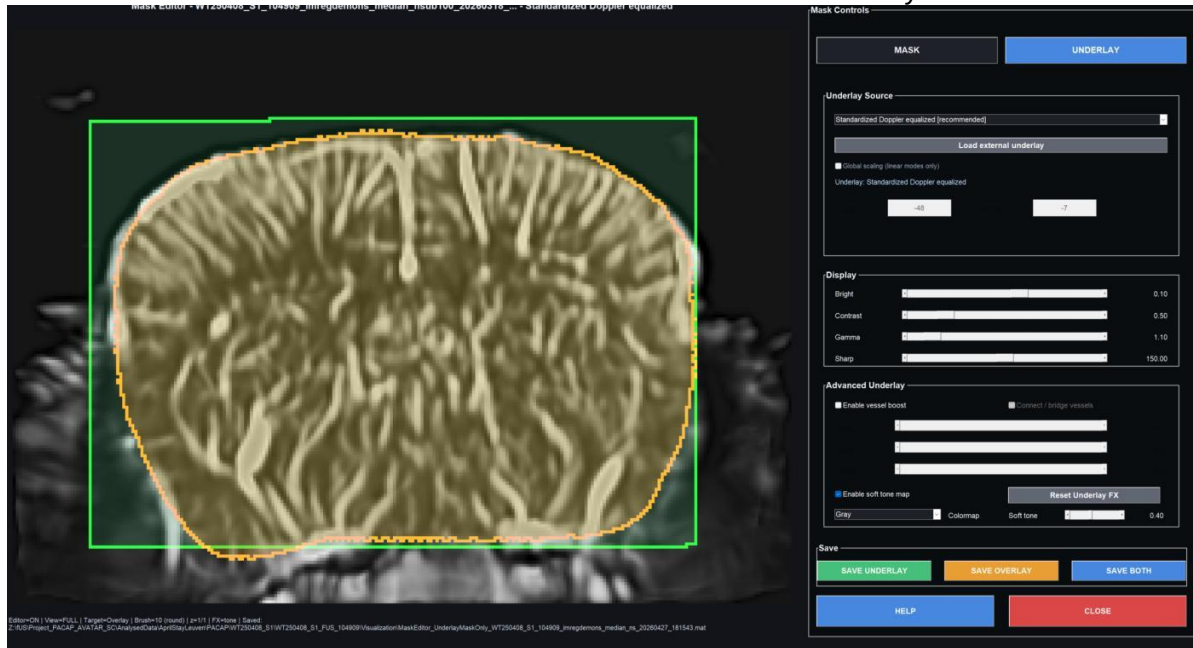


Figure 4: Mask Editor and Video GUI screenshots from SOP.

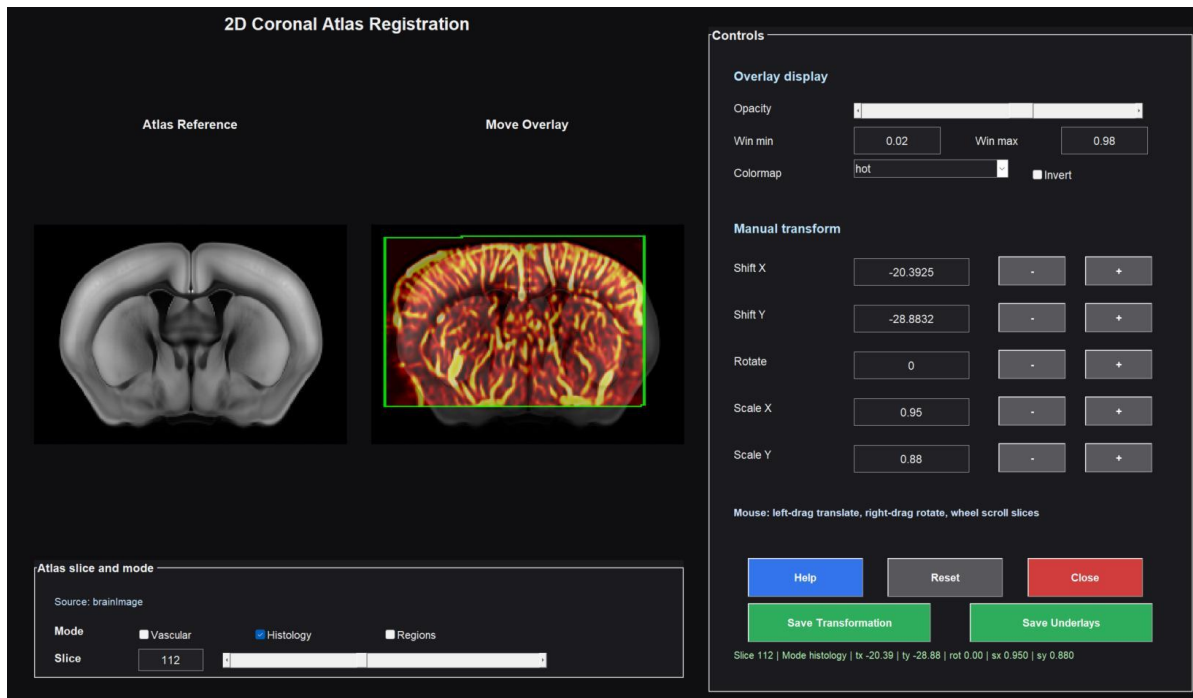


Figure 5: Atlas registration and Group Analysis screenshots from SOP.

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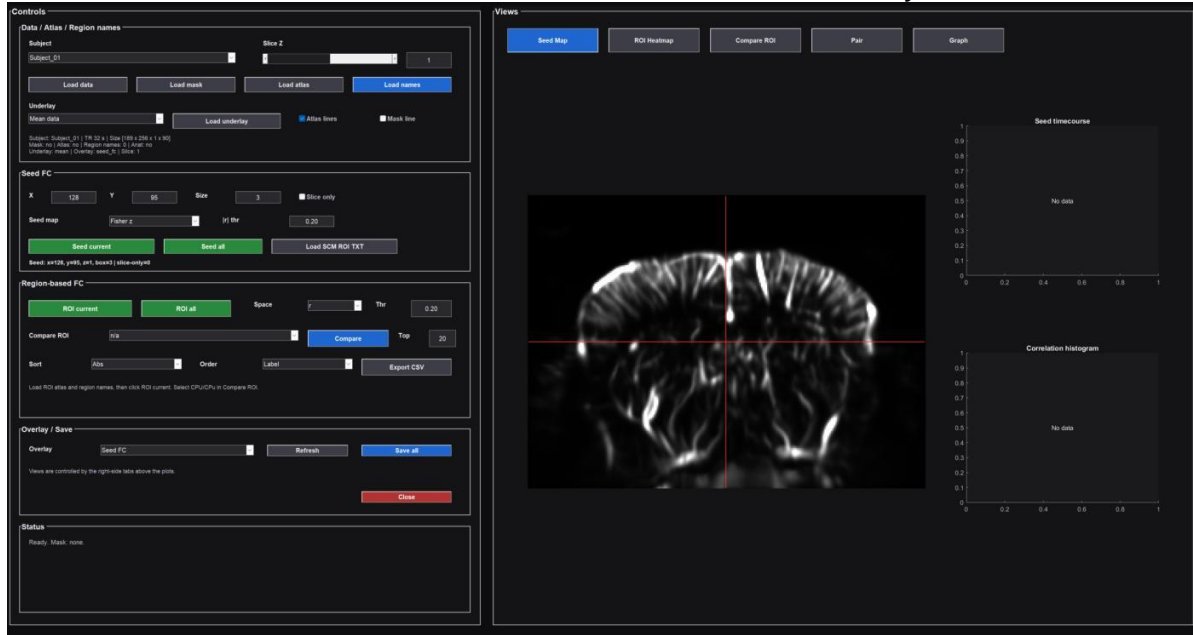


Figure 6: Functional Connectivity and Trigger Controller screenshots from SOP.

7. Detailed module guide

Button	Main code	What it does	When to use / caution
Load fUSI Data	loadFUSIData.m	Loads .mat/.nii/.nii.gz, detects imaging variable, applies Theo/pyfus geometry fixes when metadata allow, infers TR and asks user to confirm.	Always first. Verify dimensions, TR and total duration.
Dataset dropdown	fusi_studio.m state	Chooses active raw/preprocessed/PSC dataset from memory or saved outputs.	Check after every preprocessing step. The latest run is automatically chosen.
Full QC	qc_fusi.m	Runs frequency, spatial, temporal, motion/stability, frame-rate, PCA, burst, CNR, common-mode and reliability QC.	Run before modifying data.
Specific QC	qc_fusi.m module options	Runs selected QC modules for large/debug datasets.	Use when full QC is too slow or a specific diagnostic is needed.
Frame Rejection	frameRateQC.m + interpolateRejectedVolumes.m	Detects abnormal global-intensity frames and interpolates rejected volumes.	Use for acquisition-frame instabilities; document rejected count.
Imregdemons	gabriel_preprocess.m	Block mean/median plus non-rigid drift correction; supports slice-wise step-motor handling.	Use for slow drift/spatial instability, not for data already stable.
Scrubbing	scrubbing.m	DVARS or global-signal bad-frame detection with linear/PCHIP interpolation and QC.	Use for sudden motion/artifact frames.
Motor	motor.m	Reconstructs continuous or split motor acquisition into multi-slice/4D data.	Requires correct frames per position/slice or consistent split filenames.
Temporal Smoothing/Subsampling	temporalsmoothing.m	Sliding average or block mean/median; block mode changes TR.	Report changed TR and block size.
Filtering	filtering.m	Butterworth low/high/band-pass filtering with chunked time filtering and QC.	Only when frequency band is justified.
PCA / ICA	pca_denoise.m / ica_denoise.m	Interactive component selection and reconstruction after removing selected components.	Never remove components without inspecting maps/time courses.
Despike	despike.m	Voxel-wise MAD robust z-score spike detection and interpolation.	Use sparingly for sparse spikes.
Time-Course Viewer	fUSI_Live_Studio.m	Raw/PSC inspection, movie playback and quick signal overview.	Use before SCM to understand dynamics.

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SCM	computePSC.m + SCM_gui.m	Computes/displays PSC maps, ROI time courses, masks, underlays and exports.	Main individual map/ROI analysis module.
Video & SCM Mask	play_fusi_video_final.m	Dynamic playback with overlay/underlay/mask controls.	Validate that maps are temporally plausible.
Mask Editor	mask.m + makeBrushMask.m	Draws brain/underlay and overlay/signal masks; saves mask bundles.	Use to remove non-brain, bubbles or irrelevant regions.
Registration to Atlas	coreg.m, coreg_coronal_2d.m, coreg_3d.m	Manual 2D or 3D atlas registration with Allen atlas resources.	Trust atlas labels only after visual QC.
Segmentation	fusi_studio.m placeholder + segmentation helpers	Currently not the primary wired workflow.	Do not rely on it until callback integration is complete.
Functional Connectivity	FunctionalConnectivity.m	Seed/ROI/pair/graph connectivity using Pearson correlation.	Use after QC, masking and confound review.
Group Analysis	GroupAnalysis.m + modules	Organizes groups/conditions, ROI time courses, map bundles, FC bundles, outliers and exports.	Use only after individual datasets are accepted.

7.1 Quality-control interpretation

QC module	Output	Interpretation
Frequency QC	Power spectrum including broad and low-frequency ranges.	Noise rhythms, drift, acquisition periodicity.
Spatial QC	Mean image, temporal CV, tSNR map/histogram.	Vessel visibility and spatial SNR.
Temporal QC	Global signal, relative global signal, DVARS and spikes.	Abrupt jumps, drift, motion-related events.
Motion QC	Center-of-mass drift.	Apparent tissue/brain movement over time.
Stability QC	Intensity distribution, thresholding, rejected-images map.	Stable acquisition support across frames/slices.
Frame-rate QC	Global normalized intensity and rejected volume trace.	Acquisition or PNS-like instabilities.
Burst / CNR / common-mode QC	Burst ratio, pseudo/true CNR, block correlation.	Saturation-like events, response contrast and global artifacts.
Reliability QC	Finite/non-NaN support map and summary.	Whether the dataset has valid support across regions/time.

8. Group-level analysis workflow

Group Analysis is the final summary module. The uploaded code shows a modular GroupAnalysis architecture: GroupAnalysis.m as the main GUI, with backend dispatchers in GroupAnalysis_Map.m, GroupAnalysis_FC.m and GroupAnalysis_Common.m.

1. Prepare individual exports first: ROI time-course text files, SCM group bundles, masks, registration outputs or FC bundles as appropriate.
2. Open Group Analysis from HUMoR Studio only after the active individual dataset has passed QC.
3. Add subjects/animals and verify group/condition labels manually.
4. For ROI time-course analysis, load ROI exports and verify time axis, group assignment and baseline/signal windows.
5. For group-map analysis, load exported SCM group bundles rather than raw loaded Studio data.
6. Use outlier tools for inspection, not automatic deletion. Document any excluded animal/session.
7. Export Excel summaries, figures, PPT outputs and the group-analysis session state.

Tip / Trick: Before group analysis, make a simple tracking table or PPT with every animal/session, acquisition mode, preprocessing steps, exclusions and final exported file paths. This prevents accidentally mixing raw, preprocessed and PSC versions.

9. Functional connectivity workflow

FunctionalConnectivity.m provides seed, ROI heatmap, compare ROI, pair and graph-style modes based on Pearson correlation. Use it after motion/QC review and after masking/registration decisions are made.

8. Choose the active dataset version carefully. FC is sensitive to preprocessing, filtering and global artifacts.
9. Load or draw a mask to restrict analysis to valid brain/signal voxels.
10. For atlas/ROI FC, load the atlas labels and region-name information that match the registered data.
11. Select seed region or ROI pair and compute correlation.
12. Inspect whether correlations are spatially plausible or dominated by global signal/motion.
13. Export heatmaps/graphs only after confirming the time course and mask are appropriate.

$$r = \text{cov}(x,y) / (\text{std}(x) \times \text{std}(y))$$

10. Output files and naming

Folder	Typical contents	User action
QC/	QC PNGs/MAT summaries	Review before accepting dataset; useful for publication supplement/debugging.
Preprocessing/	newData MAT files from frame rejection, scrubbing, smoothing, filtering, PCA/ICA, despiking, motor/imregdemons	Each processing step should create a new named dataset.
PSC/	PSC MAT files	Signal-change data used by SCM/video workflows.
Visualization/	SCM figures, ROI exports, videos, PPTs, underlays	Main individual-level outputs.
Masks/	Brain/underlay masks, overlay/signal masks, bundles	Use clear filenames; keep with corresponding dataset.
Registration/	Transformation.mat and 3D registration resources	Dataset-specific, not source-code repository content.
Registration2D/	CoronalRegistration2D*.mat and atlas underlays	Used for 2D atlas alignment and group map warping.
FunctionalConnectivity/	FC maps, heatmaps, graphs, bundles	Only after QC/masking.
GroupAnalysis/	Excel summaries, figures, PPT outputs, group bundles	Final cohort-level outputs.

11. Quality-control checklist before publication-ready marking

QC item	Acceptable result	Action if failed
Raw movie viewed	No severe bubble/shadow/motion artifact; vascular structure visible.	Reacquire or exclude if severe.
TR confirmed	Total time matches acquisition log and stimulus/injection timing.	Reload and correct TR before further analysis.
Baseline stable	No motion, injection artifact or drift during baseline.	Move baseline or reject dataset if no clean baseline exists.
Frame outliers reviewed	Outlier count is small and interpolation is justified.	Document rejection/scrubbing settings.
Preprocessing justified	Every processing step has a QC reason.	Avoid inconsistent use across animals.

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SCM inspected	Map is anatomically and temporally plausible, not edge/bubble/global artifact.	Verify with video and ROI trace.
Mask/underlay verified	Mask aligns to the displayed slice and underlay; no wrong-slice underlay.	Reload or regenerate mask/underlay.
Registration inspected	Atlas/underlay match visually.	Do not trust atlas segmentation if registration is poor.
Group labels checked	Animal/session/condition labels correct.	Fix before statistics or export.
Studio Log exported	Analysis decisions are saved.	Export before closing session.

12. Troubleshooting

Problem	Likely cause	Fix
Dataset does not load	Unsupported variable name, corrupt MAT/NIfTI, huge file, missing reader, wrong slash/path.	Use whos/load to inspect variables. Ensure a valid 3D/4D image stack exists.
Wrong total time	Incorrect TR or suspicious metadata.	Reload and confirm TR from acquisition log or nblocksImage.
Buttons remain disabled	Data not loaded or GUI state not updated.	Load fUSI data again and check Studio Log for errors.
SCM map looks wrong	Bad baseline, wrong active dataset, motion, wrong underlay or mask issue.	Check dataset dropdown, baseline, raw movie and ROI trace.
Mask loads but wrong slice appears	Mask/underlay bundle field mismatch or slice handling mismatch.	Verify mask bundle fields and load underlay per slice; compare Mask Editor and SCM behavior.
imregdemons fails	Image too small for pyramid levels or invalid values.	Use fewer pyramid levels or skip imregdemons for small/unstable dimensions.
Motor reconstruction wrong	Wrong frames per slice/position or inconsistent split filenames.	Verify frame counts and slice_t naming before reconstruction.
PCA/ICA removes response	Biological response selected as artifact component.	Return to pre-PCA/ICA dataset and rerun with fewer/no components.
FC looks globally high	Global signal, motion or non-brain voxels dominate.	Use mask, QC and consider confound strategy before interpreting FC.
Group analysis opens old behavior	Old backup files on MATLAB path.	Keep only current modular GroupAnalysis.m and backend modules on active path.

13. Atlas

MAT file	Size	Readable metadata
allen_brain_atlas.mat	69.1 MB	Histology: (264, 160, 228), uint8; Lines: struct fields ['Cor', 'Sag', 'Tra']; Regions: (264, 160, 228), int16; Vascular: (264, 160, 228), float32; VoxelSize: (3,), uint8; infoRegions: struct fields ['acr', 'name', 'rgb', 'vol']

MAT read result: The atlas MAT file is readable as a MATLAB v5 file. It contains an atlas struct with Histology, Vascular, Regions, Lines, VoxelSize and infoRegions fields. Histology and Regions are 264 x 160 x 228 arrays; VoxelSize is 50 micrometers in all three dimensions.

Appendix A. MATLAB file inventory

This inventory is derived from the uploaded ZIP. Active/optional files are listed first; legacy/backup files should be kept outside the active MATLAB path unless intentionally debugging. Status of 30th of April 2026.

File	Category	Lines	Purpose from header comments
FunctionalConnectivity.m	Advanced analysis	3608	FunctionalConnectivity.m fUSI Studio - Functional Connectivity GUI MATLAB 2017b compatible, ASCII-only. Updated Soner/HUMoR version Fixes included in this full copy-paste version: 1) ROI heatmap is larger. 2) Graph matrix heatmap is...
GroupAnalysis.m	Advanced analysis	3541	GroupAnalysis.m Reduced modular main GUI for fUSI Studio Group Analysis MATLAB 2017b + 2023b compatible This file is intentionally smaller. Backend functions are delegated to: GroupAnalysis_Map.m GroupAnalysis_FC.m GroupAnalysis_Common.m...
GroupAnalysis_Common.m	Advanced analysis	4675	GroupAnalysis_Common - dynamic dispatcher for split GroupAnalysis module. Safe dispatcher header repaired after line-26 quote error. MATLAB 2017b + 2023b compatible. Print full module errors in Command Window. Collect all function names...
GroupAnalysis_FC.m	Advanced analysis	4919	GroupAnalysis_FC - dynamic dispatcher for split GroupAnalysis module. Safe dispatcher header repaired after line-26 quote error. MATLAB 2017b + 2023b compatible. Print full module errors in Command Window. Collect all function names in...
GroupAnalysis_Map.m	Advanced analysis	624	GroupAnalysis_Map - self-contained map backend for modular GroupAnalysis. MATLAB 2017b + 2023b compatible.
SCM_gui.m	Visualization / masks / colormaps	6501	SCM_gui - fUSI Studio SCM viewer/controller MATLAB 2017b + 2023b compatible, ASCII-safe. Corrected robust version for copy-paste use. Main fixes included: 1) Removed duplicate safeCdBack nested function conflict. 2) Robust...
apply_coronal_registration_to_map.m	Registration / atlas	168	apply_coronal_registration_to_map.m Apply saved 2D coronal registration to a 2D map or YxXt data Inputs: mapData : 2D matrix or 3D YxXt matrix regInput : Reg2D struct OR path to CoronalRegistration2D.mat atlasInput: optional atlas...
apply_reg2d_to_stack.m	Registration / atlas	49	apply_reg2d_to_stack Applies a saved 2D affine registration to a 2D image or 3D stack [Y X T] INPUT I : [Y X] or [Y X T] Reg2D : struct with fields .A and .outputSize OUTPUT J : warped image/stack in atlas space, single precision...
autoThreshold.m	Visualization / masks / colormaps	59	autoThreshold Automatic vessel-like thresholding for fUSI masks. INPUT sig : 2D signal image (PSC frame) mode : 2 = Method A (robust MAD) 3 = Method B (percentile-based) pKeep : percentile slider value (used for Method B) OUTPUT BW : ...
blackbdy_iso.m	Visualization / masks / colormaps	286	blackbdy_iso FSLeys "blackbdy_iso" colormap (0..1 RGB) Usage: cm = blackbdy_iso(); % 256 cm = blackbdy_iso(n); % resampled to n rows MATLAB 2017b compatible.
cleanupMask.m	Visualization / masks / colormaps	39	cleanupMask Remove small disconnected components from binary mask. Toolbox-safe: works with or without Image Processing Toolbox. INPUT BW : logical mask minPix : minimum cluster size (default = 15) OUTPUT BW : cleaned logical mask LOGIC:...
computeBackground.m	Visualization / masks / colormaps	28	computeBackground Compute log-compressed background anatomy for visualization. INPUT I : [nz x nx x nVols] raw fUSI data OUTPUT bg : [nz x nx] background image (double) LOGIC: IDENTICAL to fusi_video_soner10 Author: Soner Caner Cagun...
computeGabrielSignalChange.m	Visualization / masks / colormaps	77	computeGabrielSignalChange Gabriel-style signal change computation Supports: 3D input: [Y X T] 4D input: [Y X Z T] (step motor / multi-slice) OUTPUT proc.PSC : percent signal change proc.bg : background anatomy proc.nVols ---- Fixed...
computePSC.m	Core GUI / loader / paths	266	computePSC (UPDATED: supports 2D + matrix probe 3D) Classic PSC pipeline (no Gabriel branches). Supports input: - 2D probe: I = [Y X T] - Matrix probe: I = [Y X Z T] Steps: 1) temporal interpolation (integer factor N) -> preserves Z 2)...
coreg.m	Registration / atlas	851	coreg.m Unified atlas-registration launcher Updated robust version - MATLAB 2017b + 2023b compatible - ASCII-safe - Pure MATLAB source selector (no Java list dependency) - Launches Simple 2D coronal or Complex 3D atlas registration...
coreg_3d.m	Registration / atlas	782	fUSI Studio - Atlas Coregistration ASCII only MATLAB 2017b compatible Main behavior: - Lets you select an anatomy/anatomical reference file (for example your BrainOnly_*.mat with brainImage) - Opens registration GUI - Saves/loads...
coreg_coronal_2d.m	Registration / atlas	1441	coreg_coronal_2d.m Simple manual 2D coronal atlas registration launcher. Updated for 2D step-motor / multi-slice data: - No separate source-slice chooser popup is opened by default. - Full source stack is passed to...
despike.m	Preprocessing	131	VOXEL-WISE MAD DESPIKING (STABLE VERSION) STATS QC FIGURE

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draw_borders.m	Registration / atlas	36	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 Draw the border of the brain regions in coronal,...
example01_registering.m	Registration / atlas	22	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 % This section allows, with the GUI, the registration...
fUSI_Live_Studio.m	Visualization / masks / colormaps	2047	fUSI_Live_Studio Clean GUI refresh of the live viewer. Same overall functionality, improved styling/layout only. ASCII-safe. Global intensity reference Internal state Initial PSC GUI state
figviewscan.m	Utilities / optional	238	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 % FIGVIEWSCAN MATLAB code for figviewscan.fig...
filtering.m	Preprocessing	440	fUSI Studio - Advanced Butterworth Filtering Engine MATLAB 2017b + 2023b compatible Processing only. No GUI popup is created here. Recommended GUI/default settings are handled in fusi_studio: type = 'band' FcLow = 0.01 Hz FcHigh = 0.20...
frameRateQC.m	Quality control	149	frameRateQC Frame-rate / PNS QC (Urban / Montaldo style) Diagnostic only ♦ NO data modification. INPUT I : [nz x nx x nVols] fUSI data TR : repetition time (sec) tag : optional label (e.g. 'ORIGINAL', 'INTERPOLATED') savePNG : logical...
fusi_studio.m	Core GUI / loader / paths	10506	% ===== SECTION A - INTERNAL STATE & GUI CONSTRUCTION - Update FC / atlas / registration helper path % ===== FIGURE WINDOW %...
gabriel_param_gui.m	Preprocessing	47	gabriel_param_gui Parameter dialog for Gabriel preprocessing. ONLY spatial-domain parameters are allowed here. Allowed: - Gaussian spatial smoothing Disallowed (intentionally): - LPF / HPF - Temporal interpolation Author: Soner Caner...
gabriel_preprocess.m	Preprocessing	480	gabriel_preprocess Supports: - 3D input: lin [Y X T] - 4D input: lin [Y X Z T] Steps: 1) Block-averaging with MEAN or MEDIAN (user-selectable) 2) Non-rigid drift correction (imregdemons) 3) QC figures: DISPLAY (optional) + PNG export...
ica_denoise.m	Preprocessing	1096	ICA_DENOISE - interactive ICA removal (MATLAB 2017b + 2023b) Similar spirit to PCA_DENOISE, but uses: PCA reduction -> whitening -> symmetric FastICA Outputs: newData.I : denoised data stats : selection, energy proxy, QC files,...
interpolate3D.m	Registration / atlas	90	interpolate3D (ROBUST) Paper-faithful intent: - Resample scan.Data to atlas.VoxelSize - Then flip/permute axes to match atlas orientation (same as paper code) Fixes: - Avoids meshgrid/meshgridvectors issues by using ndgrid + interpn -...
interpolateRejectedVolumes.m	Quality control	41	interpolateRejectedVolumes Interpolate rejected volumes along time (Urban/Montaldo style) INPUT I : [nz x nx x nVols] outliers : logical [nVols x 1] OUTPUT Iout : same size as I LOGIC: IDENTICAL to fusi_video_soner10 Author: Soner Caner...
loadFUSIData.m	Core GUI / loader / paths	839	loadFUSIData Robust fUSI data loader with TR / time inference INPUT dataFile : full path to .mat / .nii / .nii.gz fallbackTR : fallback TR (sec) if nothing found OUTPUT data.I : [Y x X x T] or [Y x X x Z x T] single data.TR : repetition...
makeBrushMask.m	Visualization / masks / colormaps	37	makeBrushMask Create circular brush mask. INPUT x,y : center coordinates (col, row) r : radius (pixels) nz : image height nx : image width OUTPUT B : logical [nz x nx] mask LOGIC: IDENTICAL to fusi_video_soner10 Author: Soner Caner Cagun
mapscan.m	Registration / atlas	109	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 % auxiliary class to manage a 3D volume
mask.m	Visualization / masks / colormaps	3273	mask.m - fUSI Studio Mask Editor MATLAB 2017b / 2023b compatible ASCII-only source for maximum copy-paste safety PURPOSE - Draw a brain / underlay mask - Draw an overlay / signal mask - Save both into ONE MAT bundle IMPORTANT SAVE LOGIC...
motor.m	Preprocessing	1581	MOTOR RECONSTRUCTION (FRAME-BASED, TWO MODES, STUDIO-SAFE) Supports: 1) Continuous mode - One long raw movie [Y x X x T] - Motor stays N frames per slice - Optional sequential initial baseline per slice 2) Split mode - Many MAT files in...
moveimage.m	Registration / atlas	234	Simple image mover for registration GUI ASCII only MATLAB 2017b compatible Left mouse drag = translate Right mouse drag = rotate
pca_denoise.m	Preprocessing	1016	PCA_DENOISE ♦ interactive PCA removal (MATLAB 2017b + 2023b) - 25 PCs/page grid selector (dark theme) - Left-click: toggle select (red border = removed) - Right-click: deselect - Clean minutes axis ticks + labels - Logs apply/cancel...

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play_fusi_video_final.m	Visualization / masks / colormaps	6190	fUSI Video GUI (MATLAB 2023b) GUI-cleaned version ASCII only ---- defaults ---- Keep robust fallback if missing Requested overlay start range ----- DIMENSIONS (PSC) ----- UNDERLAY STATE native/original snapshots...
preproc_param_gui.m	Preprocessing	35	Simple parameter dialog for preprocessing User cancelled ? keep previous parameters Optional (safe default, ignored unless used)
preview_registered_mean_with_atlas.m	Registration / atlas	41	No header comment detected.
print_region_list.m	Registration / atlas	73	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 Generates list of brain regions 3 files with the...
qc_fusi.m	Quality control	1822	qc_fusi -- fUSI Studio QC engine (MATLAB 2017b) Keeps existing QC plots and ADDS paper-style / Python-style QC: - Burst error QC (ratio map + noisy voxels + coverage over time) - tSNR map + histogram (baseline noise), mean image,...
readFileList.m	Registration / atlas	118	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 Auxiliary function Reads a list of brain regions and...
regionGrowSimilarity.m	Visualization / masks / colormaps	85	regionGrowSimilarity Very permissive region growing based on local signal similarity. INPUT sig : 2D signal image (PSC frame) sy, sx : seed coordinates (row, col) winR : local window radius sigmaFactor : tolerance multiplier maxPix :...
register_data.m	Registration / atlas	26	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 Interpolates and registers a volumetric data with the...
registration_ccf.m	Registration / atlas	1770	Atlas registration GUI ASCII only MATLAB 2017b compatible Main goals: - Use your nice mask-editor brainImage / MIP / anatomy as overlay - Better contrast controls for overlay - Keep atlas fixed, move overlay - Save Transformation.mat -...
registration_coronal_2d.m	Registration / atlas	1733	registration_coronal_2d.m Manual simple 2D coronal atlas registration. Updated for 2D step-motor / multi-slice data: - One larger GUI. - Source slice selector is inside the main GUI. - Switch source slice with Prev/Next, slider, edit...
save_atlas_underlays_from_reg2d.m	Registration / atlas	124	save_atlas_underlays_from_reg2d Saves vascular/histology/regions atlas underlay MAT files for one Reg2D slice
scrubbing.m	Preprocessing	587	SCRUBBING (MEMORY-SAFE) - Detection: DVARS or Global Signal - Interp : Linear or PCHIP Optional trimming Full QC PNG saving MATLAB 2017b compatible ----- FORCE 4D ----- SINGLE SCRUBBING SETUP POPUP...
segment_coronal_regions_2d.m	Registration / atlas	101	segment_coronal_regions_2d.m Region-wise segmentation for 2D registered coronal data. Inputs scan2D : HxW or HxWxT numeric array atlasRegionLabelsLR2D: signed region labels negative = left hemisphere positive = right hemisphere...
segmentation_ccf.m	Registration / atlas	79	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE Review & test: C.BRUNNER, M. GRILLET September 2020 Interpolates and registers all images of the...
showScmVideoSetupDialog.m	Visualization / masks / colormaps	1024	UNDERLAY CHOICES 1 = Default (current SCM / Video bg) 2 = Mean of ACTIVE dataset 3 = Median of ACTIVE dataset 4 = External underlay file 5 = Recommended Standard (same as Mask Editor default) These are the important outputs for the caller
studio_mkdir.m	Core GUI / loader / paths	6	No header comment detected.
studio_resolve_paths.m	Core GUI / loader / paths	174	No header comment detected.
temporalsmoothing.m	Preprocessing	261	temporalsmoothing Two modes in one file: 1) Sliding temporal smoothing - centered moving average - endpoint replication - output length equals input length 2) Block averaging / subsampling - non-overlapping temporal blocks - output...
timeToVolume.m	Utilities	21	timeToVolume Convert time (seconds) to nearest valid volume index. INPUT t : time (sec) TR : repetition time (sec) nVols : total number of volumes OUTPUT volume: valid volume index (1..nVols) Author: Soner Caner Cagun Refactor: Naman Jain
tryReadSCMroiExportTxt.m	Advanced analysis	48	Tries to read SCM_gui ROI export txt: - header lines begin with '#' - table columns: time_sec time_min PSC Returns ok=false if not readable / not in expected format. detect start of numeric table
vFUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_COMMAND.m	Trigger controller	3196	ASCII safe, MATLAB 2017b compatible Backend runner used by the GUI entry file. WHAT THIS VERSION DOES 1) Keeps backward compatibility with the older config structure. 2) Supports independent enable/disable for: - StimBox - PulsePal -...

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vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_OBJECT.m	Trigger controller	773	ASCII safe MATLAB 2017b compatible Per-frame callback object for: SCAN.doppler(...,'processRF',@obj.newImage) Supports: - StimBox triggers - PulsePal triggers - GUI frame updates - user stop requests - motor scheduling inside a trial...
vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_PARAMETERS.m	Trigger controller	2687	UTF-8 safe, MATLAB 2017b compatible GUI ENTRY SCRIPT Keep this filename unchanged because the encrypted launcher may call it. Header Live measured dt/TR display. This is based on wall-clock timing between frame callback updates. Main...
volumeToFrame.m	Utilities	24	volumeToFrame Convert volume index to interpolated frame index safely. INPUT volume : 1-based volume index interpol : interpolation factor nFrames : total interpolated frames OUTPUT frame : valid frame index LOGIC: IDENTICAL to...
volumeToTime.m	Utilities	22	volumeToTime Convert volume index to time (seconds). INPUT volume : scalar or array (1-based index) TR : repetition time (sec) OUTPUT t : time in seconds LOGIC: IDENTICAL to usage in fusi_video_soner10 Author: Soner Caner Cagun Refactor:....
winter_brain_fsl.m	Visualization / masks / colormaps	285	winter_brain_fsl FSLeys-derived winter brain colormap. Returns an n x 3 RGB colormap in [0,1]. Usage: cm = winter_brain_fsl; cm = winter_brain_fsl(256);

Appendix B. Developer notes and cleanup recommendations

Topic	Recommendation
Keep active runtime path clean	Move old GroupAnalysis backups, old monoliths, parser-error versions and patch scripts into archive/ or docs/developer/ so MATLAB does not accidentally call them.
NIFTI loader guard	In loadFUSIData.m, initialize TRWasImputed = false before the switch extKey block. Currently the variable is clearly set in the .mat branch; the shared volume-consistency block uses it after the switch, which can affect .nii/.nii.gz loading.
Atlas resource management	allen_brain_atlas.mat is about 69 MB. Do not repeatedly commit it into source history; use Git LFS, a release asset or download instructions.
Generated outputs	Do not commit Preprocessing, PSC, QC, Visualization, Registration or GroupAnalysis outputs into the code repository except small curated examples. Add patterns to .gitignore.
Segmentation button	The Studio segmentation button should be documented as placeholder/experimental until the 2D/3D segmentation callback is fully wired and validated.
User release package	For users, ship a clean ZIP with core code, active modules, resources, README, example dataset and this manual. Exclude .git, backup files and generated exports.
Testing checklist	Before release, run fusi_studio from a clean MATLAB path, load representative 2D, step-motor and 3D/matrix datasets, and test each button with small data.